

EECS 348 - Assignment 7

Requirements Artifact

Functional Requirements (FR)

- FR1: The system will retrieve all students majoring in 'IS'.
- FR2: The system will find the names of students who have enrolled in more than two courses.
- FR3: The system will list professors who have taught more than 5 years in the Physics department.
- FR4: The system will retrieve the total number of students enrolled in each department where the count exceeds 50.
- FR5: The system will find course names and IDs where the title includes 'Data' and the instructor is Dr. Johnson.
- FR6: The system will display students who have not enrolled in any courses in the past two semesters.
- FR7: The system will retrieve the second-highest GPA among all students.
- FR8: The system will find names of students who are also teaching assistants and have GPA above 3.5.
- FR9: The system will list students and their courses if they enrolled after 2022.
- FR10: The system will retrieve names and salaries of the top three highest-paid professors.
- FR11: The system will insert a new Junior CS student named Alice Smith with GPA 3.85 and show her record.
- FR12: The system will update Bob Norbert's city and ZIP to Overland Park, KS, and display the result.

Non-Functional Requirements (NFR)

- NFR1: The system will follow a modular structure with separate functions for each query.
- NFR2: The program will use the MySQL Connector/C++ library for database access.
- NFR3: Each SQL query will be commented and documented in C++ for readability.

Constraints (C)

- C1: The solution must be implemented in C++ using standard libraries and MySQL Connector/C++.
- C2: Only the lab or cycle servers can be used due to MySQL access restrictions.
- C3: The database structure must match the Lab8_SQL_tables.zip format.

Assumptions (A)

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- A1: The database has already been populated with the required five tables.
- A2: Student records are up-to-date and have valid GPAs and enrollment dates.
- A3: Users running the program have valid credentials and server access.